

Convergence of Series

Problem 1

Study the convergence of the series

$$\sum_{n=1}^{\infty} \frac{n+2^n}{n2^n}.$$

Solution.

Asymptotically, we have

$$\frac{n+2^n}{n2^n} \sim \frac{1}{n}.$$

To see this, we can explicitly compute the limit of the ratio,

$$\lim_{n \rightarrow \infty} \frac{n(n+2^n)}{n2^n} = 1 + \lim_{n \rightarrow \infty} \frac{n}{2^n} = 1.$$

Since the harmonic series diverges by the integral comparison test, so does the series $\sum_{n=1}^{\infty} \frac{n+2^n}{n2^n}$. ■

Problem 2

Study the convergence of the series

$$\sum_{n=1}^{\infty} \frac{1}{e^n}.$$

Solution.

Introduce the function $f(x) = e^{-x}$ in such a way that the general term of the series is $f(n)$. Notice that f is continuous, positive and decreasing on the positive half line. The integral comparison test implies that the series $\sum_{n=1}^{\infty} \frac{1}{e^n}$ converges if and only if the improper integral $\int_1^{\infty} f(x)dx$ does. It is readily verified that $\int_1^{\infty} e^{-x}dx = e^{-1}$ so the series is convergent. ■

Problem 3

Study the convergence of the series

$$\sum_{n=1}^{\infty} \frac{n+1}{2n+3}.$$

Solution.

Observe that

$$\lim_{n \rightarrow \infty} \frac{n+1}{2n+3} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{2 + \frac{3}{n}} = \frac{1}{2} \neq 0$$

so the series diverges by the divergence test. ■

Problem 4

Study the convergence of the series

$$\sum_{n=1}^{\infty} \frac{n5^n}{10^{n+1}}.$$

Solution.

To simplify notation, write $a_n = \frac{n5^n}{10^{n+1}}$ for the general term of the series. It is clear that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)5^{n+1}}{10^{n+2}} \cdot \frac{10^{n+1}}{n5^n} = \frac{5}{10} \lim_{n \rightarrow \infty} \frac{n+1}{n} = \frac{1}{2} < 1$$

so the series $\sum_{n=1}^{\infty} a_n$ converges by the ratio test. ■

Problem 5

Study the convergence of the series

$$\sum_{n=2}^{\infty} \frac{n^2 2^n}{(2n-1)!}.$$

Solution.

To simplify notation, write $a_n = \frac{n^2 2^n}{(2n-1)!}$ for the general term of the series. Observe that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)^2 2^{n+1}}{(2(n+1)-1)!} \cdot \frac{(2n-1)!}{n^2 2^n} = 2 \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^2 \cdot \frac{(2n-1)!}{(2n+1)!}$$

and

$$\frac{(2n-1)!}{(2n+1)!} = \frac{(2n-1)!}{(2n+1) \cdot (2n) \cdot (2n-1)!} = \frac{1}{2n(2n+1)}.$$

This means that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 2 \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^2 \cdot \frac{1}{2n(2n+1)} = 0$$

so the series $\sum_{n=1}^{\infty} a_n$ converges by the ratio test. ■